WORKED SOLUTION

STEP I 2014, Question 11

Mechanics • Connected Pulley Systems

STEP I 2014, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q11

The key idea

Write Newton's second law for each particle, using one tension per string, and get the accelerations of all the particles from the string constraints. In System II the important fact is that the pulley P_1 is *light*: the net force on it must be zero (otherwise it would have infinite acceleration), so the upper string's tension is exactly twice the lower one's. Once that is in place, P_1 and its two particles act on M just as a single mass 4μ would.

Solution

Throughout, the particles are modelled as point masses, the strings as light and inextensible, and the pulleys as smooth and light, so the tension is the same all along each string.

Part (i)

Let the tension in the string be T. Since the string is inextensible, if M accelerates downwards at a_1 then m accelerates upwards at a_1. Newton's second law for each particle, taking the direction of motion as positive, gives

$$\begin{aligned} M: &\quad Mg - T = Ma_1, \\ m: &\quad T - mg = ma_1. \end{aligned}$$

Adding, $(M-m)g = (M+m)a_1$, so

$$a_1 = \frac{M-m}{M+m}\,g,$$

as required. Substituting back into the equation for m,

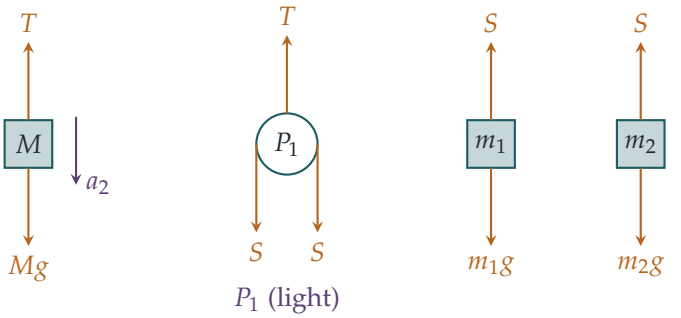
$$T = m(g + a_1) = m \cdot \frac{2M}{M+m}\,g = \frac{2Mmg}{M+m}.$$

Both straight parts of the string leave the pulley vertically downwards and each pulls with tension T, so the force on the pulley due to the tension is

$$2T = \frac{4Mmg}{M+m} \quad \text{vertically downwards.}$$

Part (ii)

Let T be the tension in the string over the fixed pulley P and S the tension in the string over P_1.



The light pulley. The forces on P_1 are T upwards and $2S$ downwards (its own weight is zero). Since its mass is zero, $T - 2S = 0 \cdot (\text{acceleration}) = 0$, so

$$T = 2S.$$

Ask yourself How do the accelerations of m_1 and m_2 relate to that of M?

The upper string is inextensible, so if M accelerates downwards at a_2 then P_1 accelerates *upwards* at a_2. The lower string only fixes the motion of m_1 and m_2 *relative to* P_1: if m_1 accelerates downwards at b relative to P_1, then m_2 accelerates upwards at b relative to P_1. Adding on the motion of P_1, the actual accelerations are

$$m_1 : \ b - a_2 \text{ downwards}, \qquad m_2 : \ b + a_2 \text{ upwards}.$$

Here b is a second unknown, which we will eliminate.

Equations of motion.

$$\begin{aligned} M : &\quad Mg - 2S = Ma_2, \\ m_1 : &\quad m_1 g - S = m_1(b - a_2), \\ m_2 : &\quad S - m_2 g = m_2(b + a_2). \end{aligned}$$

Divide the m_1 equation by m_1 and the m_2 equation by m_2:

$$g - \frac{S}{m_1} + a_2 = b = \frac{S}{m_2} - g - a_2.$$

Hence

$$2(g+a_2) = S\left(\frac{1}{m_1}+\frac{1}{m_2}\right) = S\,\frac{m_1+m_2}{m_1m_2} = \frac{S}{\mu},$$

so $S = 2\mu(g+a_2)$ and $T = 2S = 4\mu(g+a_2)$. Substituting into the equation for M,

$$Mg - 4\mu g - 4\mu a_2 = Ma_2 \quad\Longrightarrow\quad (M+4\mu)a_2 = (M-4\mu)g,$$

and therefore

$$a_2 = \frac{M-4\mu}{M+4\mu}\,g,$$

as required. Compare this with part (i): M behaves exactly as if it were hanging against a single particle of mass 4μ.

When are a_1 and a_2 equal? Put $f(x) = \dfrac{M-x}{M+x}$ for $x > 0$, so that $a_1 = f(m)\,g$ and $a_2 = f(4\mu)\,g$. Since

$$f(x) = \frac{2M}{M+x} - 1$$

and $M > 0$, f is strictly decreasing, hence one-to-one, on $x > 0$. So $a_1 = a_2$ if and only if $m = 4\mu$. With $m = m_1 + m_2$ this becomes

$$m_1+m_2 = \frac{4m_1m_2}{m_1+m_2} \iff (m_1+m_2)^2 = 4m_1m_2 \iff (m_1-m_2)^2 = 0 \iff m_1 = m_2,$$

where the first step multiplies by $m_1+m_2 > 0$, which is reversible. This completes the proof in both directions.

Checks

- If $m_1 = m_2$, the lower part of System II balances, so $b = 0$: P_1 and its particles move as one body of mass m_1+m_2, and System II reduces to System I. That matches $4\mu = m_1+m_2$ in this case.
- If $m_2 \to 0$ then $\mu \to 0$ and $a_2 \to g$: nothing heavy restrains M, since m_1 simply falls and takes no tension. Sensible.
- The tension $T = 4\mu(g+a_2)$ in part (ii) can also be found from part (i): if P_1 were fixed, the force on it would be $4m_1m_2g/(m_1+m_2) = 4\mu g$. Accelerating upwards at a_2 just replaces g by $g+a_2$.
- Numerically, with $M = 3$, $m_1 = 1$, $m_2 = 2$ we get $\mu = \frac{2}{3}$ and $a_2 = \frac{1}{17}g \approx 0.576\ \mathrm{m\,s^{-2}}$ (for $g = 9.8\ \mathrm{m\,s^{-2}}$). Solving the four equations directly gives the same value.

Common slips

- Forgetting that P_1 moves, and writing the accelerations of m_1 and m_2 as $\pm b$ instead of $b - a_2$ and $b + a_2$.

- Using $T = S$ across the light pulley. Each side of P_1 pulls down with S, so the string above it must carry $2S$.
- Concluding $a_1 = a_2 \Rightarrow m = 4\mu$ without saying why: it needs f to be one-to-one.

Beyond STEP

This set-up is an Atwood machine within an Atwood machine, and $\mu = m_1 m_2/(m_1 + m_2)$ is the *reduced mass*, the same quantity that appears when a two-body problem such as a planet and its star is reduced to a single body. Systems with constraints like these are handled most cleanly by Lagrangian mechanics, which works directly with the independent coordinates (here the height of P_1 and the position of m_1 relative to it) and never needs the tensions. Kleppner and Kolenkow, *An Introduction to Mechanics*, works through pulley systems carefully and goes on to reduced mass in the two-body problem.

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